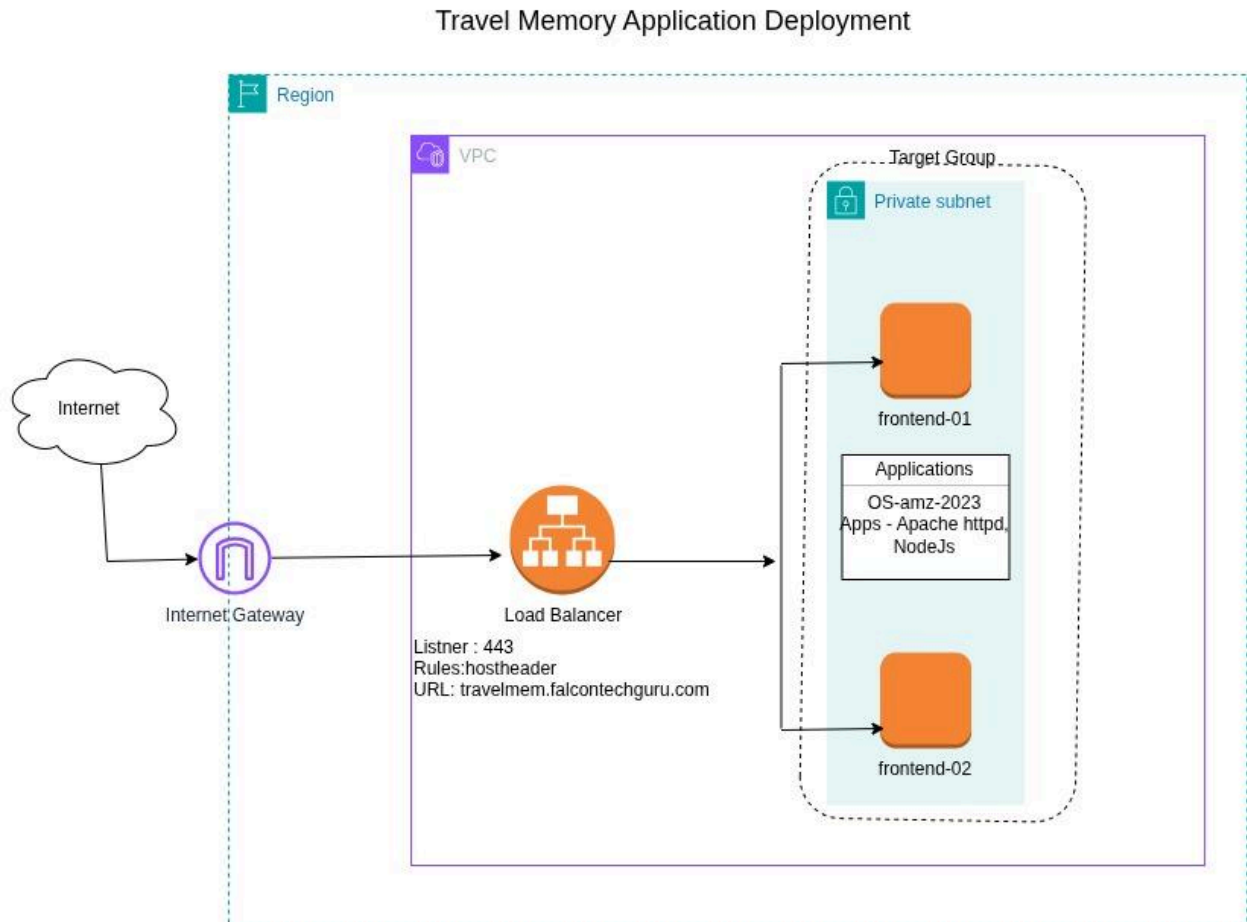


AWS Deployment Architecture Diagram.



Project Name - TravelMemory AWS Deployment (EC2 + Docker + Nginx + ALB + Cloudflare)

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Batch - 16A

Date - May 2026

Project Github Link -

<https://github.com/saleeminfodevops/herovired/tree/e0247c25ca8b45e53edf242e36e3e2e28007bfed/project/aws-deployment>

1. Created 2 Instances in ap-south-1

```
testing@inblr1t-002:~/Downloads/hv/project/fbs$ aws ec2 describe-instances --filters "Name=platform-details,Values=Linux/UNIX" --output table
DescribeInstances
+-----+-----+-----+-----+-----+-----+
| InstanceID | Name | Platform | PrivateAddress | PublicAddress | State |
+-----+-----+-----+-----+-----+-----+
| i-037f542b93fe8accd | frontend-01 | Linux/UNIX | 172.31.12.201 | 43.204.218.148 | running |
| i-0400124bba6643b1a | frontend-02 | Linux/UNIX | 172.31.13.171 | None | stopped |
+-----+-----+-----+-----+-----+-----+
```

2. Created Security Group

```
testing@inblr1t-002:~/Downloads/hv/project/fbs$ aws ec2 describe-security-groups --group-ids sg-00cd4cadf40faf3cf
{
  "SecurityGroups": [
    {
      "Description": "launch-wizard-1 created 2026-05-03T10:09:36.914Z",
      "GroupName": "launch-wizard-1",
      "IpPermissions": [
        {
          "FromPort": 80,
          "IpProtocol": "tcp",
          "IpRanges": [
            {
              "CidrIp": "172.31.0.0/16",
              "Description": "VPN CIDR"
            },
            {
              "CidrIp": "103.5.133.53/32",
              "Description": "my ip"
            }
          ]
        },
        {
          "FromPort": 22,
          "IpProtocol": "tcp",
          "IpRanges": [
            {
              "CidrIp": "103.5.133.53/32",
              "Description": "my ip"
            }
          ]
        }
      ],
      "OwnerId": "135808938060",
      "GroupId": "sg-00cd4cadf40faf3cf",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ]
        }
      ],
      "VpcId": "vpc-04db9c37f3ade1fdc"
    }
  ]
}
```

3. Installed and configured nginx for reverse proxy

```
[root@ip-172-31-12-201 ec2-user]# cat /etc/nginx/conf.d/ftg_travelmem.conf
server {
    listen 80;
    server_name travelmemory.falcontechguru.com;

    location / {
        proxy_pass http://127.0.0.1:3000; # Address of your backend app
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}

[root@ip-172-31-12-201 ec2-user]# sudo systemctl restart nginx
[root@ip-172-31-12-201 ec2-user]# sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-05-08 17:41:25 UTC; 6s ago
     Process: 154566 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 154567 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 154568 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
  Main PID: 154569 (nginx)
    Tasks: 2 (limit: 1120)
   Memory: 2.9M
      CPU: 48ms
   CGroup: /system.slice/nginx.service
           └─154569 "nginx: master process /usr/sbin/nginx"
             └─154570 "nginx: worker process"

May 08 17:41:25 ip-172-31-12-201.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
May 08 17:41:25 ip-172-31-12-201.ap-south-1.compute.internal nginx[154567]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
May 08 17:41:25 ip-172-31-12-201.ap-south-1.compute.internal nginx[154567]: nginx: configuration file /etc/nginx/nginx.conf test is successful
May 08 17:41:25 ip-172-31-12-201.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-12-201 ec2-user]#
```

5. Clone the TravelMemory repo

```
[ec2-user@ip-172-31-12-201 ~]$ git clone https://github.com/UnpredictablePrashant/TravelMemory.git
Cloning into 'TravelMemory'...
remote: Enumerating objects: 119, done.
remote: Counting objects: 100% (70/70), done.
remote: Compressing objects: 100% (51/51), done.
remote: Total 119 (delta 33), reused 19 (delta 19), pack-reused 49 (from 1)
Receiving objects: 100% (119/119), 198.68 KiB | 6.62 MiB/s, done.
Resolving deltas: 100% (40/40), done.
[ec2-user@ip-172-31-12-201 ~]$ ls TravelMemory/
LICENSE  README.md  azure-pipelines.yml  backend  frontend
[ec2-user@ip-172-31-12-201 ~]$
```

6. Created and edited the Docker config files

```
[ec2-user@ip-172-31-12-201 TravelMemory]$ cat backend/.env
MONGO_URI='mongodb+srv://ftgdbadmin:50ShYPJ2mqKqFup7@cluster0.kjbshiw.mongodb.net/?appName=Cluster0'
PORT=3000
[ec2-user@ip-172-31-12-201 TravelMemory]$ cat backend/Dockerfile
FROM node:25-alpine3.22
WORKDIR /app
COPY . .
RUN npm install
ENV PORT=3000
ENV MONGO_URI=""
EXPOSE 3000
CMD [ "node", "index.js", "url.js" ]

[ec2-user@ip-172-31-12-201 TravelMemory]$ cat frontend/src/url.js
export const baseUrl = process.env.REACT_APP_BACKEND_URL || "http://localhost:3000";
[ec2-user@ip-172-31-12-201 TravelMemory]$
```

```
[ec2-user@ip-172-31-12-201 backend]$ docker build . -t saleem-project-travelmemory
[+] Building 1.3s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 241B
=> [internal] load metadata for docker.io/library/node:25-alpine3.22
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/node:25-alpine3.22@sha256:d4f742bdae57b81d8b1a1c78e210e5c3c2b3892b335a30575956819028c804e
=> [internal] load build context
=> => transferring context: 50.53kB
=> CACHED [2/4] WORKDIR /app
=> CACHED [3/4] COPY . .
=> CACHED [4/4] RUN npm install
=> exporting to image
=> => exporting layers
=> => writing image sha256:f1f92b37ea6eb43599c1ddc104bd82122da9d03167e96c633cfc09d257a710c0
=> => naming to docker.io/library/saleem-project-travelmemory
[ec2-user@ip-172-31-12-201 backend]$ docker run -p 3000:3000 -e MONGO_URI='mongodb+srv://ftgdbadmin:50ShYPJ2mqKqFup7@cluster0.kjbshiw.mongodb.net/?appName=Cluster0' saleem-project-travelmemory
docker: Error response from daemon: driver failed programming external connectivity on endpoint pedantic_kepler (dce302e9fa86c61fdb8575c3d5a18f0e70ff2814dfc477b3d18151b4b6a57136): Bind for 0.0.0.0:3000 failed: port is already allocated.
[ec2-user@ip-172-31-12-201 backend]$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
82ca7bc8a29b   saleem-project-travelmemory         "docker-entrypoint.s..." 4 days ago    Up 4 days    0.0.0.0:3000->3000/tcp, :::3000->3000/tcp   great_lamport
[ec2-user@ip-172-31-12-201 backend]$
```

7. Build and run the TravelMemory project

```
[ec2-user@ip-172-31-12-201 backend]$ docker build . -t saleem-project-travelmemory
[+] Building 1.3s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 241B
=> [internal] load metadata for docker.io/library/node:25-alpine3.22
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/node:25-alpine3.22@sha256:d4f742bdae57b81d8b1a1c78e210e5c3c2b3892b335a30575956819028c804e
=> [internal] load build context
=> => transferring context: 50.53kB
=> CACHED [2/4] WORKDIR /app
=> CACHED [3/4] COPY . .
=> CACHED [4/4] RUN npm install
=> exporting to image
=> => exporting layers
=> => writing image sha256:f1f92b37ea6eb43599c1ddc104bd82122da9d03167e96c633cfc09d257a710c0
=> => naming to docker.io/library/saleem-project-travelmemory
[ec2-user@ip-172-31-12-201 backend]$ docker run -p 3000:3000 -e MONGO_URI='mongodb+srv://ftgdbadmin:50ShYPJ2mqKqFup7@cluster0.kjbshiw.mongodb.net/?appName=Cluster0' saleem-project-travelmemory
docker: Error response from daemon: driver failed programming external connectivity on endpoint pedantic_kepler (dce302e9fa86c61fdb8575c3d5a18f0e70ff2814dfc477b3d18151b4b6a57136): Bind for 0.0.0.0:3000 failed: port is already allocated.
[ec2-user@ip-172-31-12-201 backend]$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
82ca7bc8a29b   saleem-project-travelmemory         "docker-entrypoint.s..." 4 days ago    Up 4 days    0.0.0.0:3000->3000/tcp, :::3000->3000/tcp   great_lamport
[ec2-user@ip-172-31-12-201 backend]$
```

8. Created an ApplicationLoadBalance and HostHeader Rule

EC2 > Load balancers > ftg-lb > HTTP:80 listener

HTTP:80 Info

▼ Details

A listener checks for connection requests using the protocol and port that you configure. The default action and any additional rules that you create determine how the Application Load Balancer routes requests to its registered targets.

Protocol:Port

HTTP:80

Load balancer

[ftg-lb](#)

Default actions

- Forward to target group

[ftg-tg](#)

 1 (100%)

Target group stickiness: Off

Listener ARN

[arn:aws:elasticloadbalancing:ap-south-1:135808938060:listener/app/ftg-lb/f9c180112d3b5945/1d9b823642aba3a5](#)

Rules

Attributes

Tags

Listener rules (2) Info

Traffic received by the listener is routed according to the default action and any additional rules. Rules are evaluated in priority order from the lowest value to the highest value.

Q Filter rules

☐

Priority ▲

Name tag

Conditions (If)

Transforms

Actions (Then)

ARN

Actions

☐

1

-

Host header (value) =
[travelmemory.falcontechguru.in](#)

-

- Forward to target group

[ftg-tg](#)

 1 (100%)

Target group stickiness: Off

[ARN](#)

EC2 > Load balancers > ftg-lb

Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. [Learn more](#)

ftg-lb Info

▼ Details

Load balancer type

Application

Status

Active

VPC

[vpc-04db9c37f3ade1fddc](#)

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

ZP97RAFLXTNZK

Availability Zones

[subnet-0dccc4abdf46ad1](#) ap-south-1a (aps1-az1)

[subnet-074685289d0a53e5c](#) ap-south-1b (aps1-az3)

Date created

May 3, 2026, 17:58 (UTC+05:30)

Load balancer ARN

[arn:aws:elasticloadbalancing:ap-south-1:135808938060:loadbalancer/app/ftg-lb/f9c180112d3b5945](#)

DNS name Info

[ftg-lb-1242025918.ap-south-1.elb.amazonaws.com \(A Record\)](#)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1) Info

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Q Filter listeners

☐

Protocol:Port

▼

Default action

▼

Rules

▼

ARN

▼

Security policy

▼

Default SSL/TLS certificate

▼

mTLS

▼

Trust store

▼

☐

HTTP:80

- Forward to target group

[ftg-tg](#)

 1 (100%)

Target group stickiness: Off

[2 rules](#)

[ARN](#)

Not applicable

Not applicable

Not applicable

Not applicable

8. Registered Domain and Created a CNAME Record

falcontechguru.in

☆ Free

Ask AI

Configure DNS records and review [proxy status](#) of your hostnames.
[DNS documentation](#)

falcontechguru.in is **pending** until you complete the instructions on the [Overview page](#) and we are able to verify ownership. Learn more about [pending domains](#).

Recommendations

✓ Add an A, AAAA, or CNAME record for www and optionally [create a redirect rule](#) to send visitors to falcontechguru.in. [New Alert](#)

✓ Add an A, AAAA, or CNAME record pointing to the server that hosts your website or app.

✓ Add an MX record to receive email. [Set up SPF, DKIM, and DMARC records](#) to prevent email spoofing. [New Alert](#)

Hide

DNS management for **falcontechguru.in**

DNS Setup: Full [Import and Export](#) [Dashboard Display Settings](#)

Review, add, and edit DNS records. Changes propagate globally within minutes.

Add filter

Search DNS Records

Search

Add record

☐	Type	Name	Content	Proxy status	TTL	Actions
☐	CNAME	travelmemory	ftg-lb-1242025918.ap-sout...	Proxied	Auto	Edit